Justification for Accelerated Publication

This letter is a companion to the manuscript *The Coordinate-Space Geometry of the Tetrad Frame* (CQG-115428), currently under review at Classical and Quantum Gravity. That letter proved that the spatial sector of a single orthonormal tetrad produces two bounded tetrahedral regions related by spatial parity: TetA (the identity component) and TetB (the parity component). The present letter addresses the question that the companion geometry immediately raises: what do the 24 antisymmetric (torsion) components of the affine connection carry in the two-chamber structure of the tetrad frame?

The letter contains two formal results and one algebraic identity. First, a Lemma proves that a torsion-free connection is uniquely determined by the metric, which is bilinear in the frame field and therefore invariant under the spatial parity that exchanges TetA and TetB. A Proposition then proves that the tetrad equation is quadratic in the frame field, that TetA and TetB are its two roots in the parity direction, and that the torsion-free connection is root-blind while a torsionful connection is root-resolving. A Corollary establishes torsion as the necessary carrier of the TetA/TetB distinction at the connection level. Second, a Theorem proves that the torsion count 24 admits the polynomial factorisation $D \times 2K$ exclusively at $K = 4$, linking the antisymmetric count to the coupling range of a polynomial that organises every standard count of the tetrad formalism. Third, the exchange-symmetric partition derived in a supplementary note determines the TetB field fraction as $1/(x+1)$, whose ratio to the retained fraction is $1/x$, algebraically identical to the dark-to-baryon mass ratio of the cold dark matter framework.

The two arguments are independent: the parity identification requires only the companion geometry and textbook general relativity, while the polynomial organisation requires only the graded structure of the tetrad equation and the closure condition $D = K - 1$. A challenge to one argument does not affect the other. A self-contained validation script verifies every claim in both this letter and its companion, with 52 of 52 tests passed using symbolic computation.

The submission does not propose a modification of general relativity. It examines what geometric structure the 24 torsion components carry when they are retained in the connection. The result is kinematic: the partition between the two tetrahedral chambers is determined by the parity structure of the tetrad equation, not by any specific action principle, and therefore holds independently of the gravitational dynamics chosen. The constitutive partition satisfies the logistic equation, identifying it as the canonical two-sector transition law.

The result is relevant to the core readership of Classical and Quantum Gravity because it connects three areas of active research: the Christoffel/torsion decomposition of the affine connection, the Einstein-Cartan programme that retains torsion in the geometry, and the dark matter problem in galactic dynamics. The function derived here from the exchange symmetry of the tetrad frame was identified empirically by Milgrom (1983) and selected by fitting by Famaey and Binney (2005); the present derivation obtains it from the geometry of the two parity-related roots of the equation that defines the gravitational field. The algebraic identity between the torsion-sector field fraction and the cold dark matter residual follows from four lines of algebra with no halo profile and no free parameters. We believe this merits consideration as a Letter.